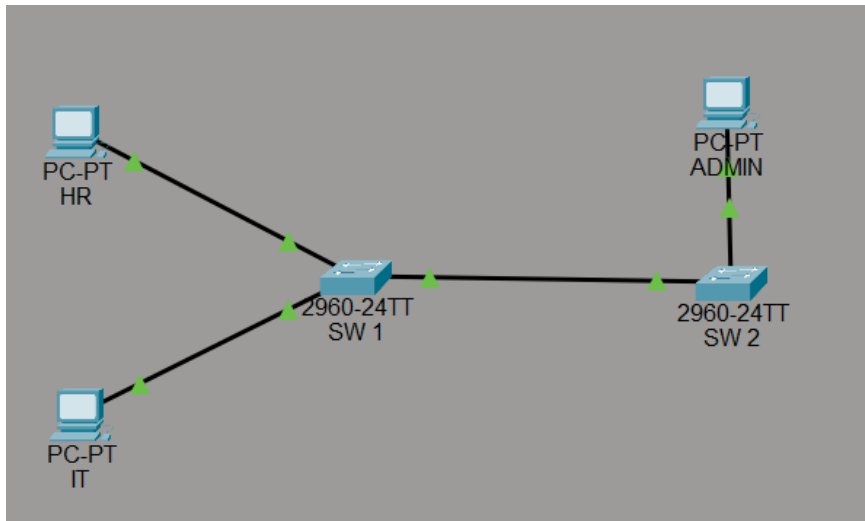


MINI PROJECT 1

VLAN + 802.1Q Trunking + Layer-2 Switching Behavior

DIAGRAM:



DESCRIPTION:

This project explains how VLANs divide a network into separate broadcast domains to reduce unnecessary traffic. It also shows how 802.1Q trunking allows multiple VLANs to travel over one link between Layer-2 switches while keeping traffic separated.

STEP 1: Create VLANs on SW 1&SW 2:

```
Enable
configure terminal
vlan 10
name HR
vlan 20
name IT
vlan 30
name ADMIN
exit
```

STEP 2: Assign Access Ports to VLANs:

SW 1,

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

```
interface fastEthernet0/2
switchport mode access
switchport access vlan 20
exit
```

SW 2,

```
interface fastEthernet0/1
switchport mode access
switchport access vlan 30
exit
```

STEP 3: Configure Trunk Link (802.1Q) on SW 1 & SW 2:

```
interface gigabitEthernet0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Subnet Mask
PC-HR	10	192.168.10.10/24	255.255.255.0
PC-IT	20	192.168.20.10/24	255.255.255.0
PC-ADMIN	30	192.168.30.10/24	255.255.255.0

Each VLAN must use a different IP network.

STEP 5: Verify VLAN & Trunk Behavior:

show vlan brief

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	HR	active	Fa0/1
20	IT	active	Fa0/2
30	ADMIN	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Show interfaces trunk

```
Switch#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig0/1	10,20,30

Port	Vlans allowed and active in management domain
Gig0/1	10,20,30

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	10,20,30

```
Switch#
```